

READING LEDGER — the atlas derivation phase (Lyra + Elie), the standard now

Every derived reading is logged here BEFORE its value is read: (object, verb, pre-registered form, target, tolerance, status, tier). Nulls sit beside hits — a sweep that reports its misses is worth more than one that doesn't. Headline of the bar: the address is named before the value.

Cal's caution (held): C6 (pre-register-or-sweep) is for a numerical pool — readings that could chance-match a target at ~1%. Do NOT apply C6 to a structural cancellation (an anomaly cancelling to 0, a projection kernel, an exact identity) — that's ritual compliance. Cancellations are tiered structurally, not swept.

Standing pins: a structural pin beats a citation — Elie pinned the Bergman genus $= n_C = 5$ via the bidisk factorization (four ways), no Hua needed, and caught the corpus's own Hua line carrying -6 . So the exponents promote without the citation. Never write " C_2/n_C " (same-number trap; $C_2=6$ and $n_C+1=6$ agree only at $n=5$). New bar criteria (now 8): C7 class-vs-object — an n -independent-within-the-class constant is a *class* property, not a reading of the object; C8 false-neighbor — $(n+1)/n$ and $n/(n-1)$ are one family at shifted arguments, so a value appearing twice in a sweep is one relation, not two maps. C6 is for numerical pools, not cancellations.

Lane A — transition maps (the atlas)

ID	object → verb	pre- registered form	target / ex- pectation	status	tier
A-1	(K_B, K_S) → kernel- exponent	disk positive- control: exponents ($n+1, n$), ratio ($n+1$)/ n at $n=1 \rightarrow 2$	ratio 2 for the disk	HIT (disk control)	control passed
A-2	(Bergman chart, Szegő chart) → Π, E	$\Pi \circ E =$ $\text{id}_{\{H^2\}}, E$ injective (reproduc- ing property T349)	proved transition on overlap $H^2 \subset A^2$	HIT — map DERIVED	DERIVED

ID	object → verb	pre- registered form	target / ex- pectation	status	tier
A-3	K_B → Bergman genus (sin- gularity exponent)	genus = n_C = 5 (bidisk, four ways)	5	HIT — DERIVED (Elie structural pin)	DERIVED (object reading)
A-4	K_S → Szegő exponent	n_C/2 = 5/2 (= ρ_1)	5/2	HIT — DERIVED	DERIVED (object reading)
A-3/4-ratio	(K_B, K_S) → exponent ratio	5 / (5/2) = 2	2	DERIVED but CLASS- LEVEL (C7) — ratio is 2 for <i>all</i> tube domains, NOT a BST number	class property, not an object reading
A-4 old	singularity ratio 5/4	n_C/(n_C-1) — = 5/4		SUPERSEDED — used wrong Szegő exp (genus-1, not n_C/2); and 6/5 & 5/4 were one false- neighbor family (C8), not two maps	DERIVED
A-5	(Cartan slice, K-type) → HC transform	Casimir $C_2(\lambda)$ = radial- Laplacian eigenvalue on A	Plancherel isomor- phism	HIT — DERIVED	DERIVED

ID	object → verb	pre- registered form	target / ex- pectation	status	tier
A-6	(K-type, Šilov stratum) → $\Pi_{\check{S}}$	$\ker \Pi_{\check{S}} = \{\lambda_2 > 0\}$; $\text{im} = \{\lambda_2 = 0\}$ (round-4 theorem)	projection with kernel = confined sector	HIT — DERIVED- structure (a <i>cancellation/kernel</i> , NOT C6-swept)	DERIVED- structure
A-N1	(bulk, Šilov stratum) → direct	a direct transition not = the Bergman ↔ Szegő composite	none expected	NULL — only closes as the composite	reported null
A-N2	(Peirce block, Cartan slice) → transition	a proved map on a chart overlap	none expected	NULL — shared base point ≠ overlap (property tested)	reported null

Lane A — confinement engine forward leads (PRE-REGISTERED, property tests, not mass computations)

ID	object → verb	pre- registered form	confirms / refutes	status	tier
A-L1	(color block V_{12} , K-type λ_2) → mem- bership	colored $\Leftrightarrow$ $\lambda_2 > 0 \Leftrightarrow$ confined	HIT — but this is T2523 (2026-07- 23), NOT new; count once. Scope (A) no free colored asymptotic states only, NOT (B) area- law/mass- gap. Criterion $\lambda_2 > 0$ (adjoint gluon (A)- confined), not N-ality. Delta = cross- check sweep 8/8 (denomi- nator stated) + Šilov- asymptotic paragraph. “T2572” RE- TRACTED (double- count); counter released.	DERIVED- structure (T2523, (A)-scope)	

ID	object → verb	pre- registered form	confirms / refutes	status	tier
A-L2	($\nu=0$ stratum, $\ker \Pi_{\check{S}}$) → mem- bership	$\nu=0$ generation $\in \ker \Pi_{\check{S}}$	refute: $\nu=0$ (tau) is color- singlet → $\lambda_2=0$ → emitted	MISS (reported) — confine- ment axis λ_2 perp generation axis ν ; owned mis-stated “lightest” ($\nu=0$ is tau)	refuted → structural clarifica- tion

Lane B — #124, what the Kostant cubic Dirac D governs (PRE-REGISTERED)

Guard: mass gate closed — no lepton re-run, no Koide. NOT Hilbert–Pólya (retracted, Elie 5286). Read the spectral structure, not a spectrum-to-mass map.

ID	object → verb	pre- registered form	target / ex- pectation	status	tier
B-1	(Kostant Dirac D) → self- adjointness	D self-adjoint in the FK metric	$\ D-D^*\ \sim$ 1e-15 (banked T2562/K1487)	HIT (banked)	DERIVED
B-2	($D \rightarrow D^2$ (Kostant/Partially formula))	$D^2 =$ $\ \mu+\rho_k\ ^2$ — $\ \mu+\rho_k\ ^2$ on G -rep V_λ , K-type μ	closed- form spectrum from Casimir difference	PRE- REGISTERED — @Elie compute	(pending)
B-3	($D \rightarrow \ker$ $D = \text{Dirac}$ cohomol- ogy)	zero modes = K-types with $\ \mu+\rho_k\ =$ $\ \lambda+\rho_g\ $ (Vogan condition)	the distin- guished K-types (Dirac co- homology)	PRE- REGISTERED — property test: does it read the spinor block / strata?	(pending, structural)

ID	object → verb	pre- registered form	target / ex- pectation	status	tier
B-4	(D) → spectral gap	first nonzero	D^2	from B-2 over the bare vacuum	a gap value (structural, not a mass)

Lyra + Elie, 2026-08-21 (reading ledger, derivation phase). The standard: (object, verb, pre-registered form, target, tolerance, status, tier); nulls beside hits; address before value; C6 for numerical pools only, NOT cancellations (Cal's caution). Lane A logged: map DERIVED (A-2), two ratios CANDIDATE pending printed Hua (A-3/A-4, never " C_2/n_C "), KAK + confinement-projection DERIVED (A-5/A-6), two reported NULLS (A-N1/A-N2), L1 HIT (color=confinement = T2523 (A)-scope, count once) beside L2 MISS (confinement axis perp generation axis, refuted). Lane B (#124) pre-registered: D self-adjoint (HIT, banked), D^2 =Kostant-Casimir-difference (B-2), $\ker D$ = Dirac cohomology (B-3), spectral gap (B-4) — mass gate closed, not Hilbert–Pólya. Nothing pushed; CP existence-only.